8.2.3 Example: Adaptive Enrichment Design

Consider the phase-3 clinical trial in severe oral mucositis introduced in Section 8.1, where two doses are compared to a placebo. Two endpoints are evaluated, and in addition to the full population, the subgroup of HPV+ patients is also considered. In total there are eight hypothesis: In the full population, H_1, H_2 denote the hypothesis corresponding to the low and high dose and H_3, H_4 the corresponding hypotheses for the secondary endpoint. For the HPV+ subgroup the hypotheses for the two doses in the primary endpoint are denoted by H_5, H_6 and for the secondary endpoint by H_7, H_8.

Figure 8.4 illustrates the stage-one graph-based testing strategy for these hypotheses. The strategy reflects that the primary objective is to test the primary endpoint for both doses in the full population, while retaining power when the effect is confined to the subpopulation. Accordingly, the initial graph allocates 70% of α to the primary hypotheses in the full population and the remaining 30% to the primary hypotheses in the HPV$^+$ population, split equally between the two doses. Upon rejection of a primary hypothesis, 40% of its local α is reallocated to the corresponding secondary endpoint. 60% of its local α is equally divided among the remaining primary hypotheses in both populations.

Let $n_{0,1}, n_{1,1}, n_{2,1}$ denote the first stage pre-planned per group sample sizes in the total population and $(m_{0,1}, m_{1,1}, m_{2,1})$ the first stage sample sizes in the HPV+ subgroup. Furthermore we, assume that a spending function $g(\gamma, t)$, as the O'Brien Fleming type spending function (8.3.15) has been specified.

Product Test Method

As combination tests for the intersection hypotheses we define the inverse normal combination function with weights $v_{J,1} = v_{J,2} = 1/\sqrt{2}$ according to the pre-planned sample sizes.

After the first stage of the trial has been performed, the first stage p-values for each of the $2^8 - 1 = 255$ intersection hypotheses $p_{J,1}, J \in \mathcal{P}(I_1)$, where $I_1 = \{1, 2, \ldots, 8\}$ can be computed as outlined in Section 7.3.2 depend on the correlation between the test statistics and wether this correlation is known.

The correlation matrix of the eight first stage Z-statistics corresponding to hypotheses $H_1, \ldots, H_8$ is derived under the assumption that all test statistics are based on normally distributed endpoint data with known common variance. Each Z-statistic compares a treatment mean to the corresponding control mean, using the appropriate sample sizes for the full population $(n_{0,1}, n_{1,1}, n_{2,1})$ and the HPV+ subgroup $(m_{0,1}, m_{1,1}, m_{2,1})$.

Two hypotheses are correlated whenever their test statistics are based on overlapping data, for example when they share the same treatment or control group, or when one population is nested within another. The covariance between two Z-statistics equals the covariance of their underlying sample mean differences, divided by the product of their standard deviations.

The correlation between test statistics of hypothesis of a primary and secondary endpoint is unknown. Hypotheses for a specific endpoint, referring to different doses within the same population are correlated only through the shared control arm, with correlations

$$\rho_1^d = \frac{1/n_{0,1}}{\sqrt{(1/n_{1,1} + 1/n_{0,1})(1/n_{2,1} + 1/n_{0,1})}},$$

$$\rho_1^{d+} = \frac{1/m_{0,1}}{\sqrt{(1/m_{1,1} + 1/m_{0,1})(1/m_{2,1} + 1/m_{0,1})}},$$

in the full and sub-population.

Across populations, correlations arise because the HPV+ subgroup is contained within the full population. For hypotheses referring to the same dose and endpoint in the two populations, the correlation for dose $i = 1, 2$

$$\rho^s_{i,1} = \sqrt{\frac{1/n_{i,1} + 1/n_{0,1}}{1/m_{i,1} + 1/m_{0,1}}}.$$

The resulting correlation matrix is given by (8.2.11).

$$\begin{array}{c|cccccccc}
 & H_1 & H_2 & H_3 & H_4 & H_5 & H_6 & H_7 & H_8 \\
\hline
H_1 & 1 & \rho^d_1 & \cdot & \cdot & \rho^s_{1,1} & \rho^d_1\rho^s_{2,1} & \cdot & \cdot \\
H_2 & & 1 & \cdot & \cdot & \rho^s_{1,1}\rho^d_1 & \rho^s_{2,1} & \cdot & \cdot \\
H_3 & & & 1 & \rho^d_1 & \cdot & \cdot & \rho^s_{1,1} & \rho^s_{2,1}\rho^d_1 \\
H_4 & & & & 1 & \cdot & \cdot & \rho^s_{1,1}\rho^d_1 & \rho^s_{2,1} \\
H_5 & & & & & 1 & \rho^{d+}_1 & \cdot & \cdot \\
H_6 & & & & & & 1 & \cdot & \cdot \\
H_7 & & & & & & & 1 & \rho^{d+}_1 \\
H_8 & & & & & & & & 1
\end{array} \qquad (8.2.11)$$

After the first stage, the first stage information fraction t is observed (which, for simplicity, we assume to be equal for all hypotheses) and the marginal first stage p-value vector $\mathbf{p_1} = (p_1, \ldots, p_8)$ is obtained and the first stage p-values $p_{J,1}$ of all intersection hypotheses are computed. The graph and the spending function determine the first stage rejection boundaries $\alpha_{J,1}$ and we reject all intersection hypotheses for which $p_{J,1} \leq \alpha_{J,1}$. Applying the closed testing principle we can then complete the first stage test for the individual null hypotheses.

Assume that no hypothesis can be rejected at interim and that there is only a weak trend in the overall population which appears to be mainly driven by the observed treatment effect in the subgroup. Therefore, it is decided to adapt the trial and to continue in the HPV+ population. In addition, the second stage sample sizes $m_{0,2}, m_{1,2}, m_{2,2}$ may be adapted based on the first stage results. To reflect these adaptations, also the graph is updated. In particular, the nodes corresponding to the hypothesis regarding the full population are dropped and the respective edges removed and the remaining edges and node weights renormalized. This results in the graph in Figure 8.5.

Now, the second stage test is performed and second stage p-values (computed from second stage observations only) $p_{1,(2)}, \ldots, p_{4,(2)}$ are observed. To compute the second stage p-values for the intersection hypotheses, we need to compute the correlation of the second stage test statistics. These are given by

	H_5	H_6	H_7	H_8
H_5	1	ρ_2^{d+}	$\cdot$	$\cdot$
H_6		1	$\cdot$	$\cdot$
H_7			1	ρ_2^{d+}
H_8				1

where ρ_2^{d+} is computed as ρ_1^{d+} but with the second stage sample sizes.

Based on this correlation matrix we can compute the secodn stage intersection test

For the numeric example, the sample size of the trial is driven by the test for the primary endpoint in the full population. With $n = 200$ per group, assuming event proportions $p_1 = 0.50$ and $p_2 = 0.33$, a two-sided significance level $\alpha = 0.00875$, and using Pearson's chi-squared test (pooled variance), a fixed sample test provides a power of 80 to detect an absolute difference of 0.17 between groups. We plan an interim analysis after 50% of the participants have been observed and use the O'Brien Fleming type spending function (8.3.15).

As a numeric example, assume that the prevalence of the subgroup is 40% such that the first stage per group sample sizes in the HPV+ population are 40 patients. Assume the individual first stage p-values are given by

$$\mathbf{p} = \big(0.312,\ 0.211,\ 0.01,\ 0.0155,\ 0.09,\ 0.06,\ 0.089,\ 0.003\big)$$

Thus, we see only a weak trend in the overall population which appears to be mainly driven by the observed treatment effect in the subgroup. Therefore, it is decided to adapt the trial and to continue in the HPV+ population. In addition the sample size is adapted and the remaining 300 patient overall are equally allocated to the

In addition the sample size is adapted and the remaining 300 patient overall are equally allocated to the

Conditional Error Based Method

We first need to define the pre-planned design, which is a group sequential design adjusting for multiple testing using a graph-based multiple testing procedure.

Let $n_0, n_1, n2$ denote the pre-planned per group sample sizes in the total population and (m_0, m_1, m_2) the sample sizes in the HPV+ subgroup.

The correlation matrix of the eight Z-statistics corresponding to hypotheses $H_1, \dots, H_8$ is derived under the assumption that all test statistics are based on normally distributed endpoint data with known common variance. Each Z-statistic compares a treatment mean to the corresponding control mean, using the appropriate sample sizes for the full population (n_0, n_1, n_2) and the HPV+ subgroup (m_0, m_1, m_2).

Two hypotheses are correlated whenever their test statistics are based on overlapping data, for example when they share the same treatment or control group, or when one population is nested within another. The covariance between two Z-statistics equals the covariance of their underlying sample mean differences, divided by the product of their standard deviations.

Within the same population and dose, the correlation between the primary and secondary endpoint hypotheses equals the endpoint correlation ρ. Hypotheses referring to different doses within the same population are correlated only through the shared control arm, with correlation

$$\frac{1/n_0}{\sqrt{(1/n_1+1/n_0)(1/n_2+1/n_0)}}.$$

Across populations, correlations arise because the HPV+ subgroup is contained within the full population. For hypotheses referring to the same dose and endpoint in the two populations, the correlation is

$$\mathrm{Corr}(H_i, H_j) = \sqrt{\frac{1/n_d+1/n_0}{1/m_d+1/m_0}}.$$

Correlations involving a primary and a secondary endpoint are further multiplied by ρ. If ρ is not defined, all entries involving primary–secondary pairs are set to `NA`.

We are primarily interested in testing the primary endpoint for both doses in the full population, but we also want to have power in settings where there is an effect only in the subpopulation. Therefore, in the initial graph, we allocate 70% of the significance level to the two primary hypotheses in the full population and the remaining 30% to the HPV+ population, splitting it equally between the two doses. If one of the primary hypotheses is rejected, the significance level is reallocated and split between the corresponding secondary endpoint and the other primary hypotheses from both populations (see Figure 8.4).

8.3 Adaptive Closed Conditional Error Based Tests for Multiple Hypotheses

We now consider an alternative two-stage adaptive procedure for testing the k initially specified null hypotheses $H_j, j = 1, \ldots, k$, indexed by $I_1 = \{1, \ldots, k\}$, based on applying the conditional error rate principle to closed testing. At the planning phase a two-stage, level-α, group sequential test φ_J, permitting early efficacy stopping at stage one, is defined for all intersection hypothesis H_J, $J \subseteq I_1$. At the end of stage one, some individual hypotheses may demonstrate early efficacy and be rejected, while some others may be retained arbitrarily

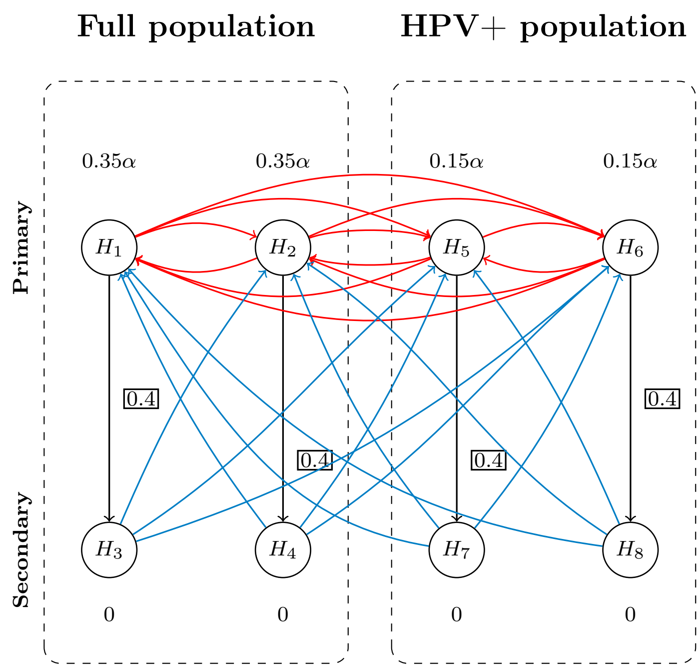

Fig. 8.4 Graph-based multiple testing procedure. All blue edges (linking the secondary endpoint hypotheses to the primary endpoint hypotheses) are assigned an edge weight of 1/3. All red edges (linking the primary endpoint hypotheses) are assigned an edge weight of 0.2.

at the behest of the trial sponsors. The remaining individual hypotheses, $H_i, i \in I_2 \subseteq I_1$, continue to accrue data and are tested at the end of stage two under the closed testing framework by intersection hypothesis tests $\tilde{\varphi}_J$ that may have undergone adaptations, such as sample size re-estimation or changes in the testing strategy, and may therefore differ from the corresponding pre-specified tests φ_J. In order to control the FWER the modified tests are are required to adhere to the Müller and Schäfer [2004] conditional error rate principle, which is the focus of this section.

For simplicity we do not consider early stopping for futility. Non-binding futility boundaries can be always be added on without any changes to these tests. While binding futility boundaries can be readily incorporated into the methods discussed here, they are almost never used in practice, for the reason discussed in comment 4 on page 204, and would thus complicate the notation unnecessarily.

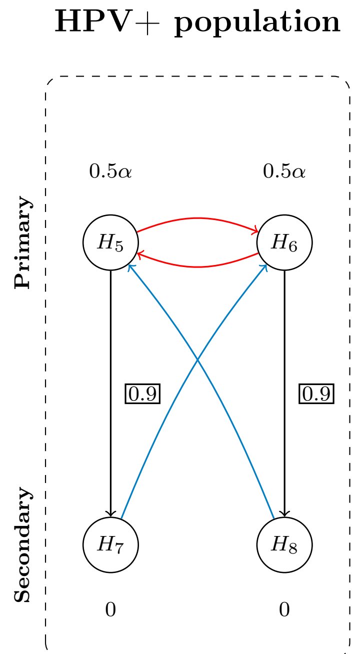

Fig. 8.5 Adapted graph-based multiple testing procedure. All blue edges (linking the secondary endpoint hypotheses to the primary endpoint hypotheses) are assigned an edge weight of 1. All red edges (linking the primary endpoint hypotheses) are assigned an edge weight of 0.4.

8.3.1 Pre-Specification of Tests for all Intersection Hypotheses

Let $P_{j,1}$ denote the p-value for testing the individual hypothesis H_j based on the data at stage one, and let $P_{j,2}$ denote the p-value for testing the individual hypothesis H_j based on the cumulative data from both stages, for all the individual hypotheses $H_j, j \in I_1$. Let $0 < w_{j,J} \leq 1$, with $\sum_{j \in J} w_{j,J} \leq 1$, be the initial weights associated with testing the intersection hypothesis H_J for all $J \subseteq I_1$, possibly elicited from a graphical testing strategy for H_{I_1} as discussed in Chapter 7. The correlation between the p-values $P_{j,1}$ and $P_{j,2}$ across stages one and two can be derived from the well-known group sequential result

$$\text{corr}(Z_{P_{j,1}}, Z_{P_{j,2}}) = \sqrt{\frac{n_{j,1}}{n_{j,2}}} \tag{8.3.12}$$

where $n_{j,1}$ is the sample size at stage one for testing H_j, $n_{j,2}$ is the initially specified cumulative sample size over both stages for testing H_j, and $Z_\gamma = \Phi^{-1}(1-\gamma)$. On the other hand the within-stage correlation $\text{corr}(P_{j,i}, P_{j',i})$